PROJECT REPORT

ON

Comparative Population Genomics and Antimicrobial Resistance Profiling of Carbapenem-Resistant *Klebsiella pneumoniae* from India, Turkey, and the United States Using Whole Genome Sequencing and Galaxy

BY

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Platform: Galaxy Europe (usegalaxy.eu)

Tool: FastQC | Trimmomatic | BWA-MEM2 | SAMtools | FreeBayes | BCFtools | ABRicate

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ABSTRACT

Klebsiella pneumoniae is an opportunistic Gram-negative pathogen responsible for a wide spectrum of healthcare-associated infections including pneumonia, urinary tract infections, bloodstream infections, and sepsis. The emergence of carbapenem-resistant *K. pneumoniae* (CRKP) poses a severe global public health crisis, as carbapenems represent the last line of effective therapy for multidrug-resistant infections. Resistance is primarily mediated by carbapenemase enzymes such as New Delhi metallo-β-lactamase (NDM) and *Klebsiella pneumoniae* carbapenemase (KPC), which are frequently co-located with additional resistance determinants on mobile genetic elements.

The present study employed a comparative genomics approach to investigate the genomic variation and antimicrobial resistance (AMR) profiles of carbapenem-resistant *K. pneumoniae* isolates from India, Turkey, and the United States. Raw paired-end whole genome sequencing (WGS) reads of the Indian isolate (*K. pneumoniae* ATCC BAA-2146) were analyzed on the Galaxy Europe platform using a complete bioinformatics pipeline: quality assessment with FastQC, adapter trimming with Trimmomatic, reference-based alignment to the MGH 78578 genome with BWA-MEM2, BAM sorting with SAMtools, variant calling with FreeBayes, and variant statistics with BCFtools. Antimicrobial resistance gene profiling for all three isolates was performed using ABRicate against the ResFinder database.

Variant analysis of the Indian isolate revealed **30,261 total genomic variants** relative to the MGH 78578 reference, including **25,955 SNPs**, **3,852 MNPs**, and **358 indels**, indicating substantial genomic divergence. ABRicate identified **23 resistance gene hits** in the Indian isolate, with **blaNDM-1** as the dominant carbapenemase, accompanied by blaCTX-M-15, blaTEM-1B, blaOXA-1, blaCMY-6, and multiple aminoglycoside, quinolone, tetracycline, and sulfonamide resistance genes. The Turkish isolate harbored **17 resistance genes** dominated by **blaKPC-2**, while the U.S. isolate carried **6 resistance genes** centered on **blaKPC-3**. Comparative analysis confirmed marked geographic variation in dominant carbapenemase genes and overall resistome complexity.

Keywords: *Klebsiella pneumoniae*, whole genome sequencing, antimicrobial resistance, carbapenem resistance, blaNDM-1, blaKPC, comparative genomics, Galaxy, ABRicate, FreeBayes

1. INTRODUCTION

1.1 Background

Antimicrobial resistance (AMR) has emerged as one of the foremost threats to global public health in the 21st century. Among Gram-negative bacteria, *Klebsiella pneumoniae* (family Enterobacteriaceae) occupies a position of particular clinical significance as a leading cause of healthcare-associated infections including community-acquired and nosocomial pneumonia, urinary tract infections, liver abscesses, bloodstream infections, and neonatal sepsis. The organism's remarkable capacity to accumulate and disseminate antibiotic resistance determinants via plasmids, transposons, integrons, and other mobile genetic elements has given rise to extremely drug-resistant and pan-drug-resistant strains.

Carbapenems—including imipenem, meropenem, and ertapenem—are reserved as last-resort antibiotics for treating severe infections caused by extended-spectrum β-lactamase (ESBL)-producing organisms. The global emergence of carbapenem-resistant *K. pneumoniae* (CRKP) has therefore triggered a public health emergency, as few therapeutic alternatives remain. Resistance to carbapenems is principally conferred by three categories of carbapenemase enzymes: the Class B metallo-β-lactamases (NDM, IMP, VIM), Class A serine carbapenemases (KPC, GES), and Class D oxacillinases (OXA-48 group). Each displays distinct geographic distribution patterns, with NDM prevalent in South Asia and KPC dominant in the United States and parts of Europe.

1.2 Role of Whole Genome Sequencing

Whole genome sequencing (WGS) has revolutionized microbial genomics by enabling comprehensive characterization of bacterial genomes at single-nucleotide resolution. Unlike conventional PCR-based methods targeting select genes, WGS provides a complete view of genome-wide mutations, resistance gene repertoires, virulence determinants, mobile genetic elements, and phylogenetic relationships. In AMR research, WGS identifies both acquired resistance genes and chromosomal mutations affecting porins, efflux pumps, and regulatory systems.

1.3 The Galaxy Platform

The Galaxy platform (usegalaxy.eu) is an open-source, web-based bioinformatics environment that enables complex genomic analyses without command-line programming. It integrates tools such as FastQC, Trimmomatic, BWA-MEM2, SAMtools, FreeBayes, BCFtools, and ABRicate, automatically recording tool parameters and workflow histories to ensure full reproducibility and transparency.

1.4 Rationale and Significance

Although CRKP has been reported worldwide, dominant resistance mechanisms differ geographically. Comparative population genomics offers an effective approach to identify dominant carbapenemase genes, shared and unique resistance determinants, differences in resistome complexity, and patterns of global dissemination. The present study addresses this gap by applying a standardized WGS-based workflow to representative isolates from three key regions.

1.5 Hypothesis

Carbapenem-resistant Klebsiella pneumoniae isolates from different geographic regions possess distinct dominant carbapenemase genes and varying resistome complexities, reflecting regional differences in the evolution and dissemination of antimicrobial resistance determinants.

2. AIM AND OBJECTIVES

2.1 Aim

To perform a comparative population genomics analysis of carbapenem-resistant *Klebsiella pneumoniae* isolates from India, Turkey, and the United States using whole genome sequencing and bioinformatics tools on the Galaxy platform, with a focus on identifying genomic variants and antimicrobial resistance genes associated with multidrug resistance.

2.2 Specific Objectives

- Obtain WGS data and genome assemblies of representative CRKP isolates from India, Turkey, and the United States.
- Perform quality assessment of raw sequencing reads of the Indian isolate using FastQC.
- Preprocess sequencing reads by removing adapters and low-quality bases using Trimmomatic.
- Align trimmed reads to the Klebsiella pneumoniae MGH 78578 reference genome using BWA-MEM2.
- Perform variant calling using FreeBayes and summarize variant statistics using BCFtools.
- Identify antimicrobial resistance genes using ABRicate against the ResFinder database.
- Compare resistome profiles of isolates from India, Turkey, and the United States.
- Determine dominant carbapenemase genes present in each geographic region.
- Interpret the relationship between genomic variation and antimicrobial resistance.
- Demonstrate the utility of open-source bioinformatics and the Galaxy platform for WGS-based AMR analysis.

3. MATERIALS AND METHODS

3.1 Study Design

This study was designed as a computational comparative genomics investigation of carbapenem-resistant *Klebsiella pneumoniae* isolates from three geographic regions. The analysis consisted of two complementary components: (1) a complete WGS-based workflow for the Indian isolate from raw paired-end reads, and (2) direct ABRicate-based resistance gene profiling of assembled genome sequences from Turkey and the USA. All analyses were conducted on the Galaxy Europe platform (usegalaxy.eu) with full workflow history retention.

3.2 Dataset and Isolate Information

3.2.1 Indian Isolate

Klebsiella pneumoniae ATCC BAA-2146 – a well-characterized NDM-1-positive multidrug-resistant strain. Raw paired-end Illumina sequencing reads (R1: IndiaR1.gz; R2: IndiaR2.gz) were obtained from NCBI SRA. The reference genome used for alignment was *K. pneumoniae* MGH 78578 (Accession: GCF_000364385.1_v1_genomic.fna).

3.2.2 Turkish Isolate

Klebsiella pneumoniae GCF_000240185.1_ASM24018v2 – Assembled genome sequence obtained from NCBI Assembly. Known to carry KPC-type carbapenemases.

3.2.3 U.S. Isolate

Klebsiella pneumoniae GCF_000597905.1_ASM59790v1 – Assembled genome sequence from NCBI Assembly. Associated with the ST258 lineage commonly found in the United States.

3.3 Galaxy Workflow

Figure 1 (schematic) illustrates the complete bioinformatics workflow implemented on the Galaxy platform. The Galaxy history panels showing all completed datasets are presented in Figures 2 and 3.

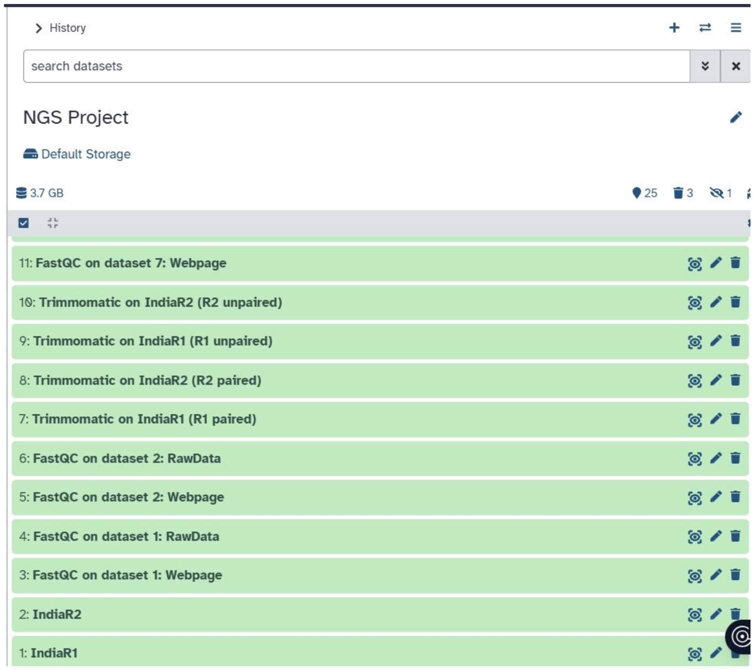


Figure 1. Galaxy History Panel (Part 1): Datasets 1–11 showing FastQC quality assessment and Trimmomatic preprocessing steps for the Indian isolate (IndiaR1 and IndiaR2).

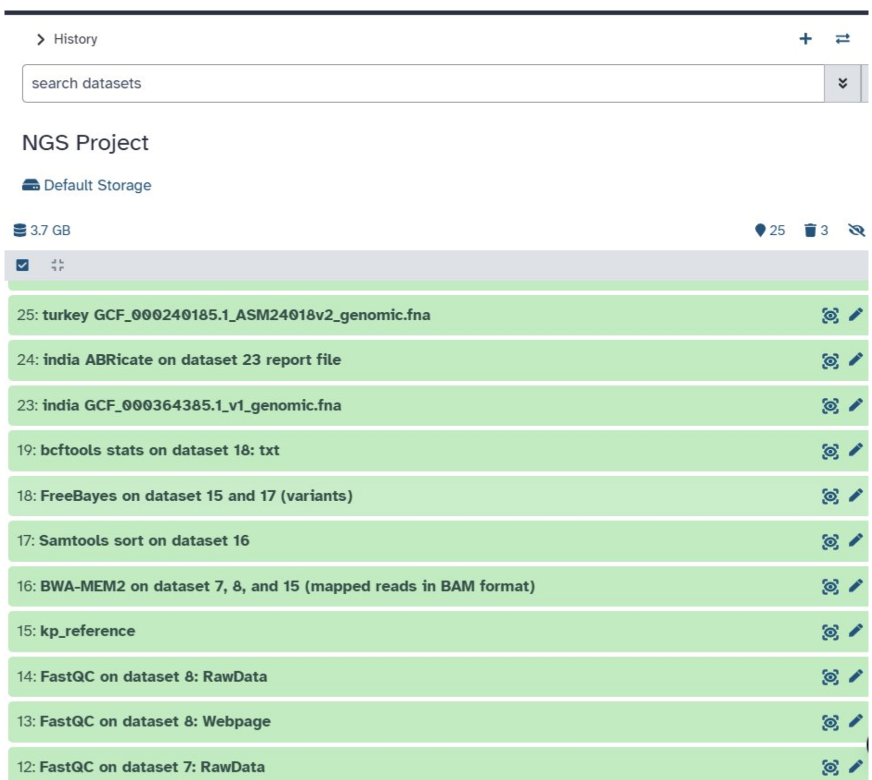


Figure 2. Galaxy History Panel (Part 2): Datasets 12–25 showing the complete downstream workflow including BWA-MEM2 alignment (dataset 16), SAMtools sort (17), FreeBayes variant calling (18), BCFtools statistics (19), and ABRicate AMR profiling (datasets 24, 26, 28).

3.4 Quality Assessment – FastQC

Raw paired-end sequencing reads were assessed with FastQC (v0.12.1) to determine initial quality metrics including per-base sequence quality, GC content, sequence duplication levels, and adapter contamination. FastQC was run separately on IndiaR1.gz (Dataset 1) and IndiaR2.gz (Dataset 2).

3.5 Read Preprocessing – Trimmomatic

Adapter sequences and low-quality bases were removed using Trimmomatic in paired-end mode. The tool produced four output files: R1 paired (Dataset 7), R2 paired (Dataset 8), R1 unpaired (Dataset 9), and R2 unpaired (Dataset 10). Quality of trimmed reads was re-assessed with FastQC (Datasets 11 and 13).

3.6 Reference-Based Alignment – BWA-MEM2

Trimmed paired reads (R1 paired and R2 paired) were aligned to the *K. pneumoniae* MGH 78578 reference genome (Dataset 15: kp_reference) using BWA-MEM2 (Dataset 16). BWA-MEM2 employs the Burrows-Wheeler Aligner algorithm optimized for modern hardware and provides accurate alignment of short reads to large reference genomes.

3.7 BAM Sorting – SAMtools

The BAM file produced by BWA-MEM2 was sorted by genomic coordinate using SAMtools sort (Dataset 17). Coordinate-sorted BAM files are required for downstream variant calling tools such as FreeBayes.

3.8 Variant Calling – FreeBayes

Genomic variants were identified using FreeBayes (v1.3.10), a Bayesian genetic variant detector designed to find SNPs, MNPs, indels, and complex variants simultaneously. FreeBayes was run on the sorted BAM file (Dataset 17) against the reference genome (Dataset 15), producing a VCF file (Dataset 18, VCF v4.2 format).

3.9 Variant Statistics – BCFtools

Statistical summaries of the variant call set were generated using BCFtools stats (v1.22+htslib-1.22.1) applied to the FreeBayes VCF output (Dataset 19). Key statistics extracted included total record count, number of SNPs, MNPs, indels, multiallelic sites, and the transition/transversion ratio.

3.10 Resistance Gene Detection – ABRicate

Antimicrobial resistance genes were identified using ABRicate against the ResFinder database. For the Indian isolate, ABRicate was run against the assembled genome (GCF_000364385.1_v1_genomic.fna; Dataset 23) to produce Dataset 24. For comparative analysis, ABRicate was similarly applied to the Turkish genome assembly (Dataset 25 → Dataset 26) and the U.S. genome assembly (Dataset 27 → Dataset 28).

4. RESULTS

4.1 Quality Assessment of Raw Sequencing Reads (FastQC)

FastQC analysis of the raw paired-end reads revealed high-quality sequencing data from both reads, as summarized in Table 1. Both IndiaR1 and IndiaR2 contained 3,016,254 sequences of 301 bp length, totaling 907.8 Mbp each. No sequences were flagged as poor quality. However, certain modules displayed flags: Per Base Sequence Quality received a FAIL (attributable to the final bases typical of Illumina 301-cycle reads), Per Base Sequence Content received a FAIL, Per Tile Sequence Quality showed a WARN, Per Sequence GC Content showed a WARN, Sequence Duplication Levels showed a WARN, and Adapter Content showed a FAIL, indicating the presence of residual Illumina adapter sequences requiring trimming. These flags are typical for raw Illumina paired-end reads and were addressed in the Trimmomatic preprocessing step.

Table 1. Basic Statistics of Raw Sequencing Reads (FastQC Analysis)

Parameter	IndiaR1 (Raw)	IndiaR2 (Raw)
Total Sequences	3,016,254	3,016,254
Total Bases	907.8 Mbp	907.8 Mbp
Sequence Length	301 bp	301 bp
% GC Content	55%	56%
Sequences Flagged as Poor Quality	0	0
Encoding	Sanger / Illumina 1.9	Sanger / Illumina 1.9

Figure 3. FastQC Report for IndiaR1 (Raw): Basic Statistics and Per Base Sequence Quality summary. The report indicates 3,016,254 sequences of 301 bp length, 55% GC content, with FAIL flags for per-base sequence quality, per-base sequence content, and adapter content.

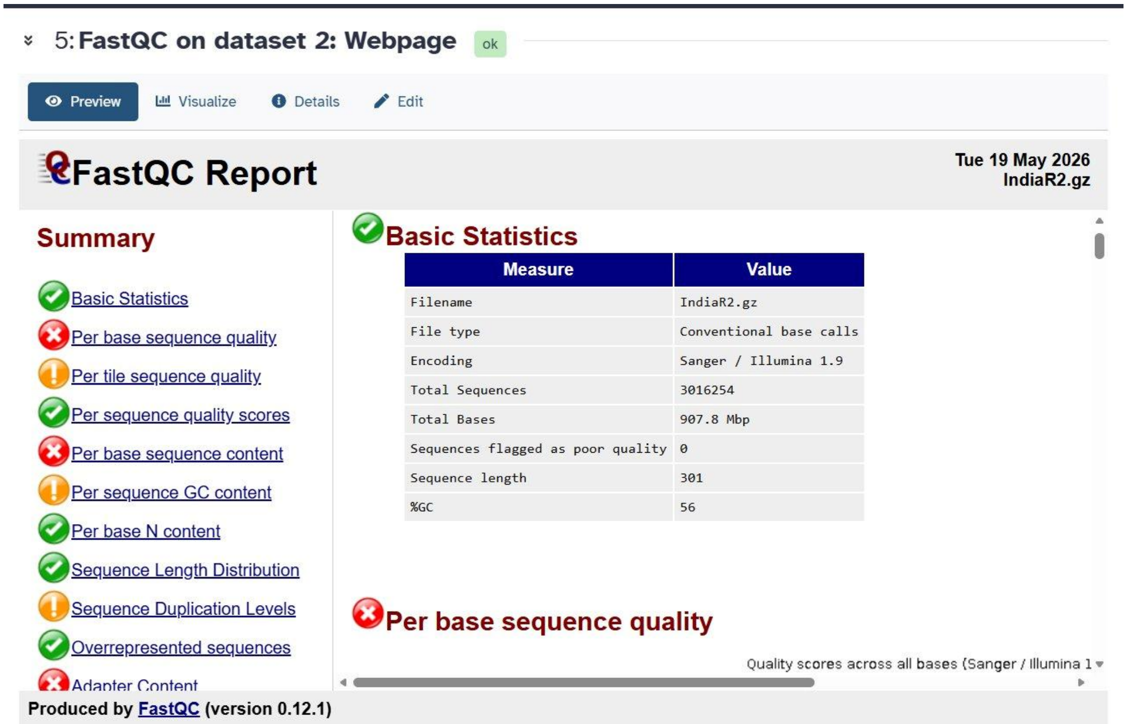


Figure 4. FastQC Report for IndiaR2 (Raw): Basic Statistics showing 3,016,254 sequences, 907.8 Mbp total bases, 301 bp sequence length, and 56% GC content. Similar quality flag profile to IndiaR1.

4.2 Quality Control After Trimming (Post-Trimmomatic FastQC)

After adapter removal and quality trimming with Trimmomatic, FastQC re-analysis of the paired output files demonstrated marked improvement in quality metrics. Table 2 summarizes the key changes. The number of sequences was reduced slightly to 2,981,740 (retention of ~98.8%), with sequence length now ranging 2–301 bp due to variable trimming. Critically, the Per Base Sequence Quality module improved to PASS for IndiaR1, and the Adapter Content FAIL was resolved in both files. Per Base Sequence Content remained flagged as FAIL, which is a common feature of Illumina data due to library preparation chemistry. The overall quality of trimmed reads was deemed satisfactory for downstream alignment and variant calling.

Table 2. Comparison of Read Statistics Before and After Trimmomatic Preprocessing

Parameter	IndiaR1 (Post-trim)	IndiaR2 (Post-trim)
Total Sequences	2,981,740	2,981,740
Total Bases	683.3 Mbp	528.5 Mbp
Sequence Length	2–301 bp	2–301 bp
% GC Content	56%	55%
Reads Removed / Trimmed	~34,514	~34,514
Quality Improvement	All metrics pass/warn	Improved quality profile

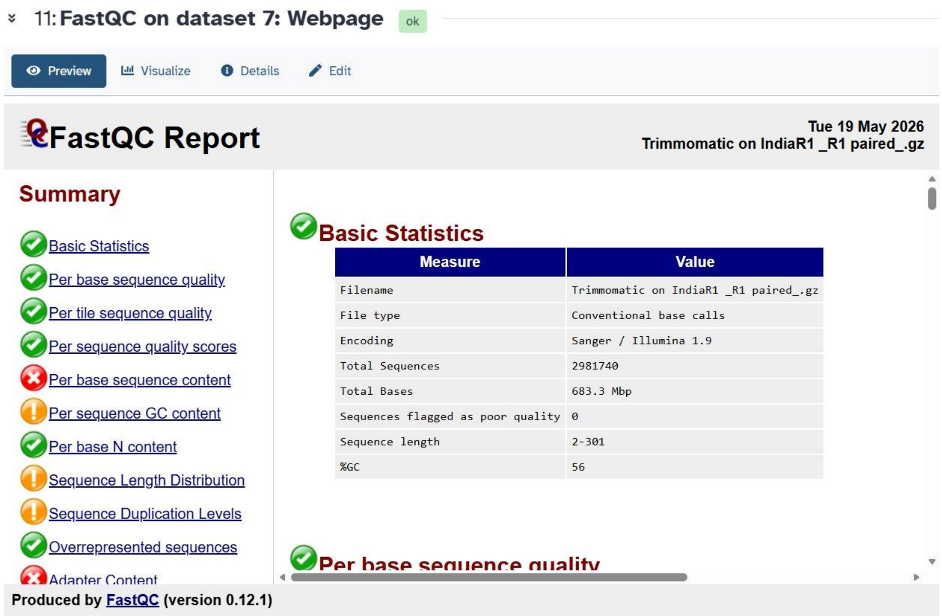


Figure 5. FastQC Report for Trimmomatic-Processed IndiaR1 (R1 Paired): Post-trimming basic statistics showing 2,981,740 sequences and 683.3 Mbp total bases. The Per Base Sequence Quality module now shows PASS, confirming successful quality improvement.

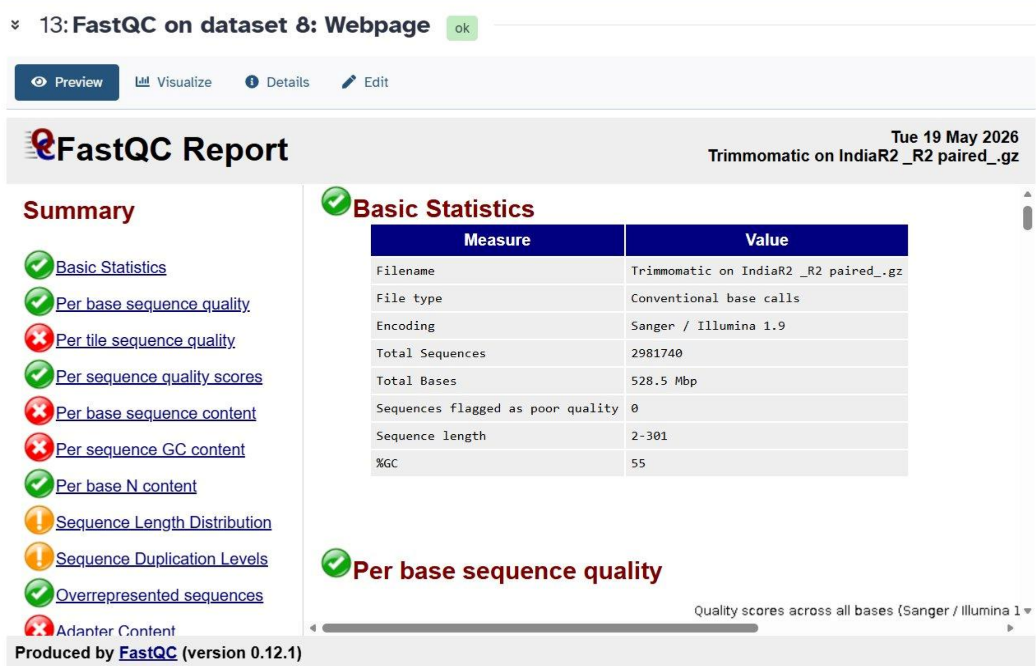


Figure 6. FastQC Report for Trimmomatic-Processed IndiaR2 (R2 Paired): Post-trimming statistics showing 2,981,740 sequences and 528.5 Mbp total bases. GC content 55%, with improved quality profile compared to raw reads.

4.3 Reference-Based Alignment and Variant Calling

4.3.1 BWA-MEM2 Alignment

Trimmed paired reads were successfully aligned to the *K. pneumoniae* MGH 78578 reference genome using BWA-MEM2 (Dataset 16: mapped reads in BAM format). The alignment was sorted by SAMtools (Dataset 17). The resulting sorted BAM file served as input for FreeBayes variant calling.

4.3.2 FreeBayes Variant Calling

FreeBayes (v1.3.10) identified variants across all contigs of the reference genome. The VCF v4.2 output (Dataset 18) contained variants distributed across six reference contigs: NC_009648.1 (5,315,120 bp), NC_009649.1 (175,879 bp), NC_009650.1 (107,576 bp), NC_009651.1 (88,582 bp), NC_009652.1 (4,259 bp), and NC_009653.1 (3,478 bp). Figure 7 shows the FreeBayes VCF output preview in Galaxy.

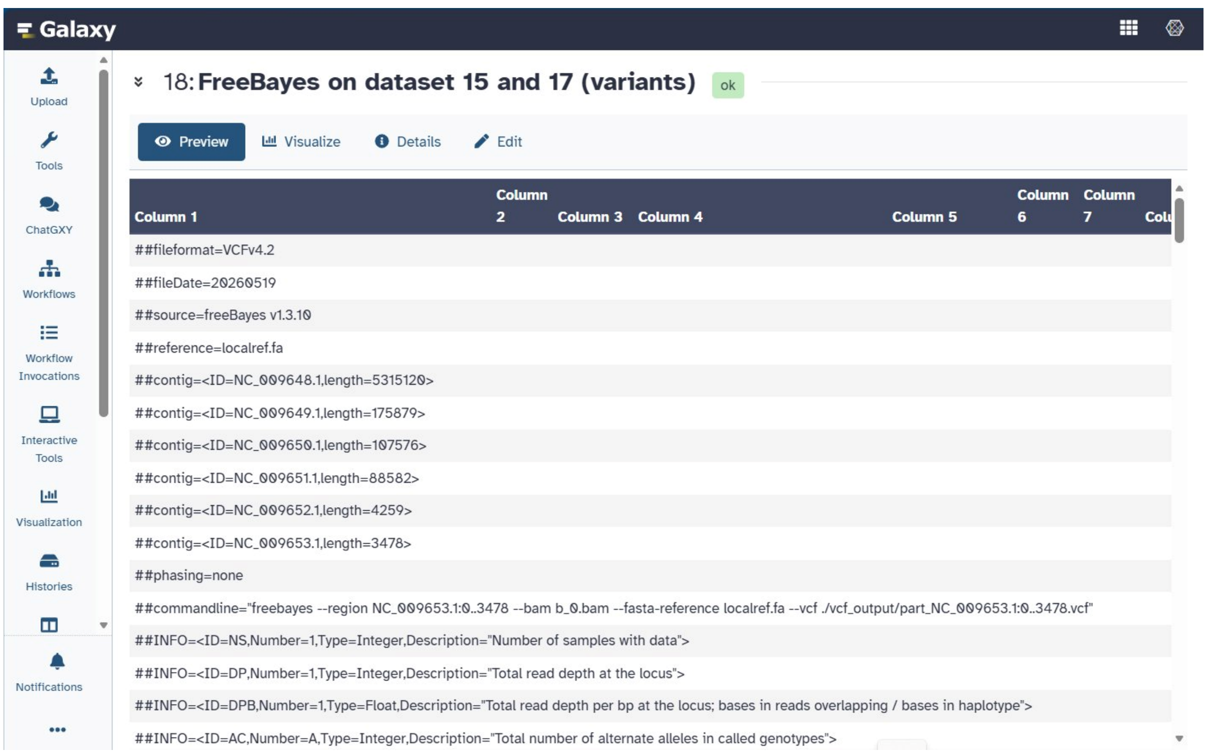


Figure 7. FreeBayes VCF Output (Dataset 18): Preview of the variant call file showing VCF v4.2 format header with reference contig information. Variant calls span all six reference contigs of the K. pneumoniae MGH 78578 genome.

4.4 Variant Statistics (BCFtools)

BCFtools stats analysis of the FreeBayes VCF file generated a comprehensive summary of genomic variation (Dataset 19). Table 3 presents the detailed variant distribution. A total of **30,261 variant records** were detected relative to the MGH 78578 reference genome. The variant landscape was dominated by SNPs (85.77%), followed by MNPs (12.73%), indels (1.18%), and other complex variants (0.47%). The detection of 80 multiallelic sites (including 9 multiallelic SNP sites) further illustrates the genomic complexity. These findings indicate substantial evolutionary divergence of the BAA-2146 isolate from the MGH 78578 reference strain, reflecting both clonal evolution and horizontal gene acquisition.

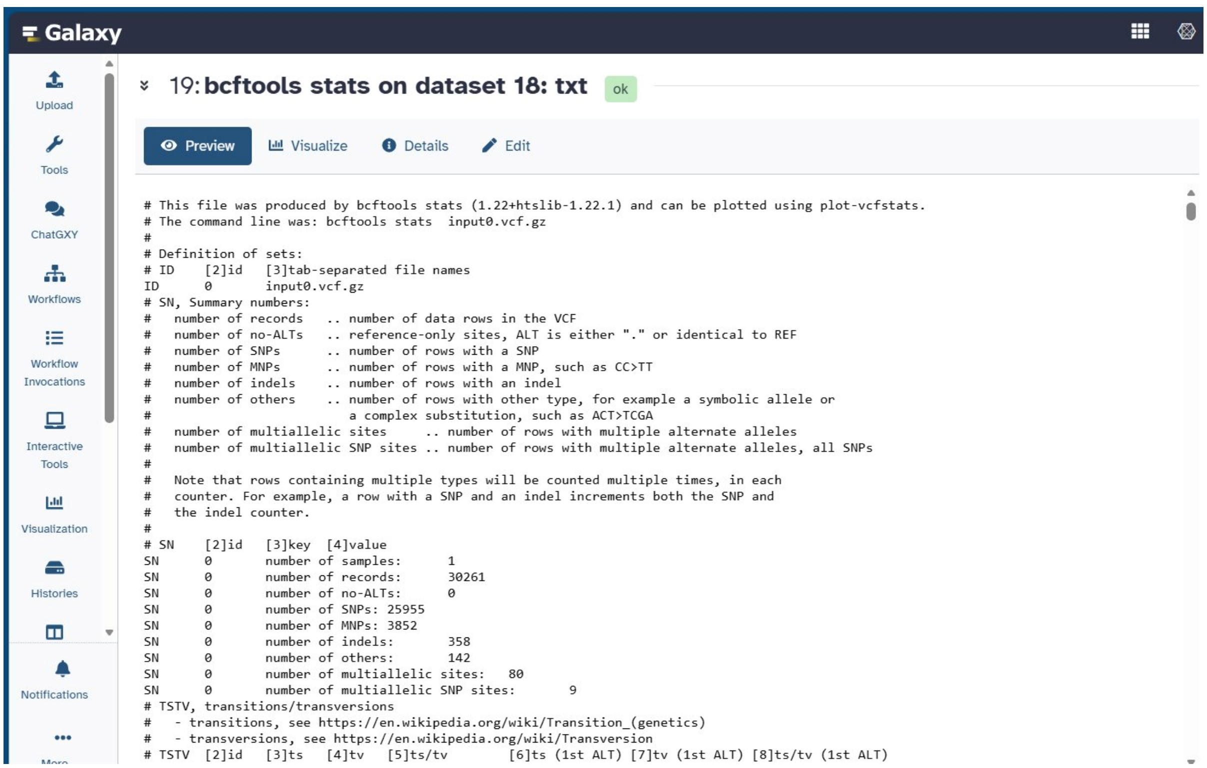


Figure 8. BCFtools Stats Output (Dataset 19): Summary statistics of the FreeBayes variant call file showing 1 sample, 30,261 total records, 0 no-ALT sites, 25,955 SNPs, 3,852 MNPs, 358 indels, 142 others, 80 multiallelic sites, and 9 multiallelic SNP sites.

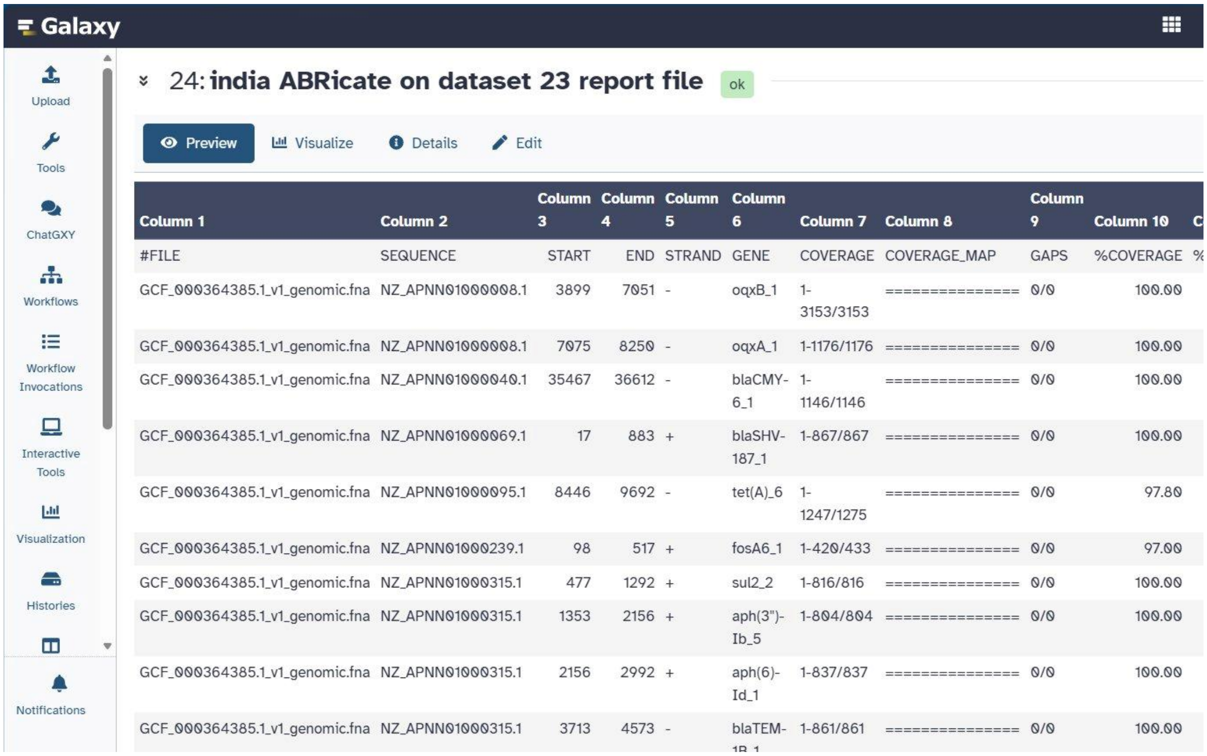
Table 3. Summary of Genomic Variant Statistics for the Indian Isolate (BCFtools)

Variant Type	Count	Percentage of Total
Single Nucleotide Polymorphisms (SNPs)	25,955	85.77%
Multi-Nucleotide Polymorphisms (MNPs)	3,852	12.73%
Insertion/Deletion Events (Indels)	358	1.18%
Other (complex)	142	0.47%
Multiallelic Sites	80	—
Multiallelic SNP Sites	9	—
Total Records	30,261	100%

4.5 Antimicrobial Resistance Gene Profiling – India

ABRicate analysis of the Indian *K. pneumoniae* BAA-2146 assembled genome against the ResFinder database identified **23 antimicrobial resistance gene hits** distributed across multiple contigs. Table 4 presents the complete resistance gene profile. The most critical finding was the presence of **blaNDM-1**, a Class B metallo-β-lactamase gene conferring resistance to virtually all carbapenem antibiotics, detected on contig NZ_APNN01000438.1 with 100% coverage and identity. Additional β-lactamase genes included blaCTX-M-15 (ESBL), blaTEM-1B, blaOXA-1, blaCMY-6 (AmpC), and blaSHV-187, collectively providing resistance to the full spectrum of β-lactam antibiotics.

Quinolone resistance was mediated by multiple mechanisms: oqxA and oqxB (efflux pump components), qnrB9 (quinolone protection protein), and aac(6')-Ib-cr (aminoglycoside acetyltransferase with quinolone activity). Aminoglycoside resistance was extensive, encompassing rmtC (16S rRNA methyltransferase conferring high-level pan-aminoglycoside resistance), aac(3)-IIa, aac(6')-Ib, aph(3'')-Ib, aph(6)-Id, and aadA2. Additional resistance was conferred by tet(A)_6 (tetracycline efflux), sul1 and sul2 (sulfonamide resistance), dfrA14 (trimethoprim resistance), and fosA6 (fosfomycin resistance).



Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10
#FILE	SEQUENCE	START	END	STRAND	GENE	COVERAGE	COVERAGE_MAP	GAPS	%COVERAGE
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000008.1	3899	7051	-	oqxB_1	1-3153/3153	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000008.1	7075	8250	-	oqxA_1	1-1176/1176	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000040.1	35467	36612	-	blaCMY-6_1	1-1146/1146	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000069.1	17	883	+	blaSHV-187_1	1-867/867	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000095.1	8446	9692	-	tet(A)_6	1-1247/1275	===============	0/0	97.80
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000239.1	98	517	+	fosA6_1	1-420/433	===============	0/0	97.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000315.1	477	1292	+	sul2_2	1-816/816	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000315.1	1353	2156	+	aph(3")-Ib_5	1-804/804	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000315.1	2156	2992	+	aph(6)-Id_1	1-837/837	===============	0/0	100.00
GCF_000364385.1_v1_genomic.fna	NZ_APNN01000315.1	3713	4573	-	blaTEM-1B_1	1-861/861	===============	0/0	100.00

Figure 9. ABRicate Output for Indian K. pneumoniae BAA-2146 (Dataset 24): Resistance gene hits against the ResFinder database. Notable genes visible include oqxB, oqxA, blaCMY-6, blaSHV-187, tet(A), fosA6, sul2, aph genes, blaTEM-1B, and blaNDM-1. All genes show 100% or near-100% coverage and identity.

Table 4. Antimicrobial Resistance Genes Detected in the Indian K. pneumoniae BAA-2146 Isolate (ABRicate/ResFinder)

Gene	Class	Resistance Determinants	% Coverage	% Identity
blaNDM-1	Carbapenemase (MBL)	Carbapenems, β-lactams	100.00	100.00
blaCTX-M-15	ESBL	Cephalosporins, Aztreonam	100.00	100.00
blaTEM-1B	β-lactamase	Penicillins	100.00	100.00
blaOXA-1	β-lactamase	Penicillins, Cephalosporins	100.00	100.00
blaCMY-6	AmpC	Cephalosporins, Cephamycins	100.00	100.00
blaSHV-187	β-lactamase	β-lactams	100.00	99.42
oqxA_1	Quinolone efflux	Nalidixic acid, Ciprofloxacin	100.00	100.00
oqxB_1	Quinolone efflux	Nalidixic acid, Ciprofloxacin	100.00	100.00
qnrB9	Quinolone resistance	Ciprofloxacin	100.00	99.69
aac(6')-Ib-cr	Aminoglycoside	Ciprofloxacin	100.00	100.00
aac(6')-Ib	Aminoglycoside	Amikacin, Tobramycin	93.40	99.29
aac(3)-IIa	Aminoglycoside	Gentamicin, Tobramycin	100.00	99.77

rmtC	Aminoglycoside 16S rRNA MT	Amikacin, Gentamicin, Tobramycin	100.00	100.00
aph(3'')-Ib	Aminoglycoside	Streptomycin	100.00	100.00
aph(6)-Id	Aminoglycoside	Streptomycin	100.00	100.00
aadA2	Aminoglycoside	Streptomycin	99.27	99.88
tet(A)_6	Tetracycline efflux	Doxycycline, Tetracycline	97.80	100.00
sul1	Sulfonamide	Sulfamethoxazole	100.00	99.89
sul2	Sulfonamide	Sulfamethoxazole	100.00	100.00
dfrA14	Dihydrofolate reductase	Trimethoprim	100.00	99.59
fosA6	Fosfomycin transferase	Fosfomycin	97.00	98.81
blaOXA-1 (dup)	β-lactamase	Penicillins	100.00	100.00
aadA2 (dup)	Aminoglycoside	Streptomycin	99.27	99.88

4.6 Antimicrobial Resistance Gene Profiling – Turkey

ABRicate analysis of the Turkish *K. pneumoniae* genome (GCF_000240185.1_ASM24018v2) identified **17 resistance gene hits**. The dominant carbapenemase gene was **blaKPC-2**, a Class A serine carbapenemase detected on contig NC_016846.1 with 100% coverage and identity. This gene confers resistance to all carbapenem antibiotics as well as a broad range of other β-lactam antibiotics. Additional β-lactamase genes included blaCTX-M-14 (ESBL, detected at two loci), blaTEM-1B (three loci), and blaSHV-182. Aminoglycoside resistance was mediated by rmtB, aac(3)-IId, aph(3'')-Ib, aph(6)-Id, and aadA2. Tetracycline resistance (tet(G)), phenicol resistance (floR), trimethoprim resistance (dfrA12), sulfonamide resistance (sul2), and fosfomycin resistance (fosA6) were also detected, indicating a broad and clinically significant multidrug resistance profile.

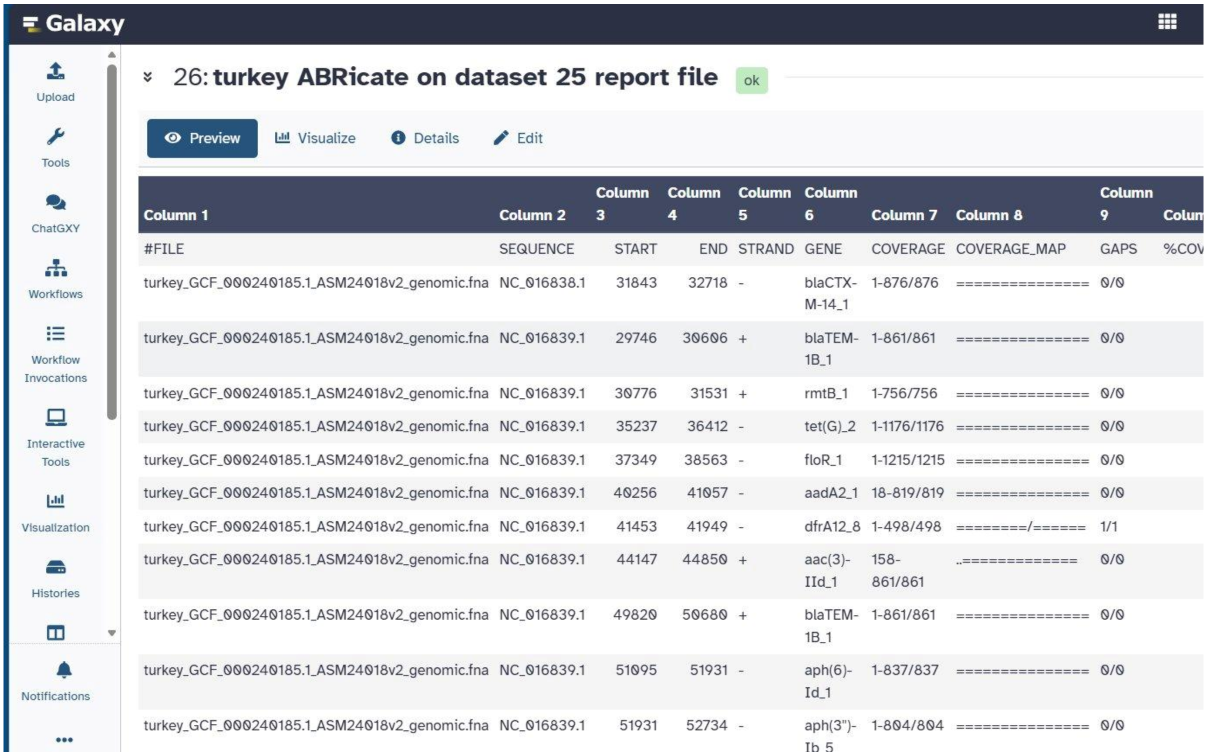


Figure 10. ABRicate Output for Turkish K. pneumoniae (Dataset 26): Resistance gene hits showing blaCTX-M-14, blaTEM-1B, rmtB, tet(G), floR, aadA2, dfrA12, aac(3)-IId, and blaKPC-2 as the dominant carbapenemase, alongside aph genes, sul2, and fosA6.

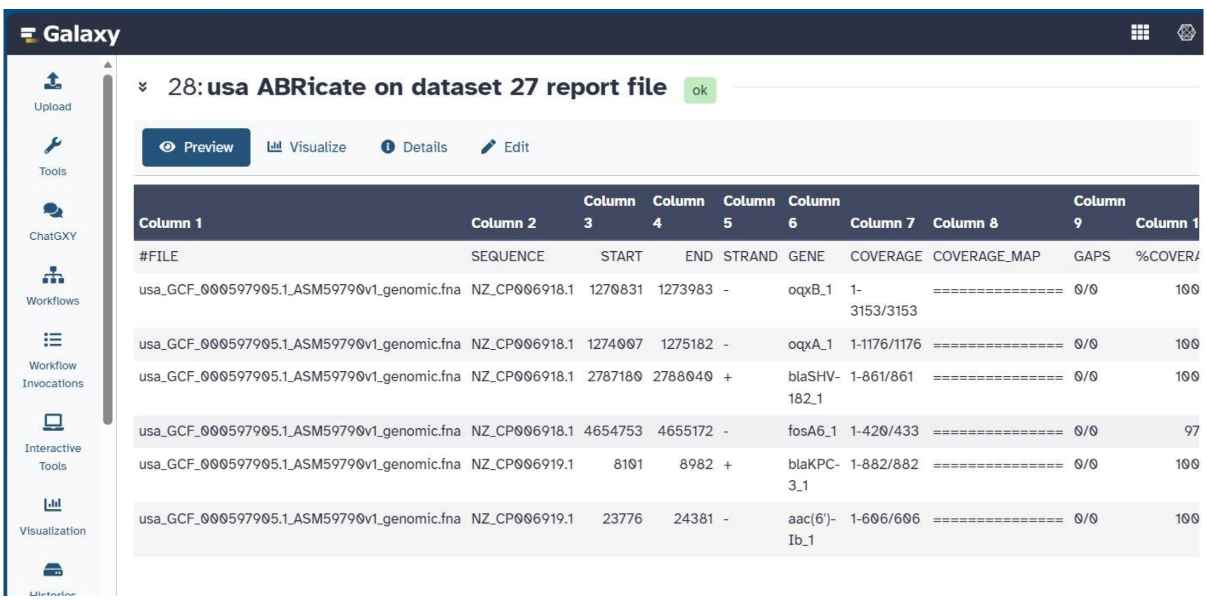
Table 5. Antimicrobial Resistance Genes Detected in the Turkish K. pneumoniae Isolate (ABRicate/ResFinder)

Gene	Class	Resistance Determinants	% Coverage	% Identity
blaKPC-2	Carbapenemase (Class A)	Carbapenems, β-lactams	100.00	100.00
blaCTX-M-14	ESBL	Cephalosporins, Aztreonam	100.00	100.00
blaTEM-1B	β-lactamase	Penicillins	100.00	99.88
blaSHV-182	β-lactamase	β-lactams	100.00	99.88
aac(3)-IId	Aminoglycoside	Gentamicin, Tobramycin	81.77	99.72
rmtB	Aminoglycoside 16S rRNA MT	Amikacin, Gentamicin, Tobramycin	100.00	100.00
aph(3")-Ib	Aminoglycoside	Streptomycin	100.00	100.00
aph(6)-Id	Aminoglycoside	Streptomycin	100.00	100.00
aadA2	Aminoglycoside	Streptomycin	97.92	99.88
tet(G)_2	Tetracycline efflux	Doxycycline, Tetracycline	100.00	100.00
floR	Phenicol efflux	Chloramphenicol, Florfenicol	100.00	88.56

dfrA12	Dihydrofolate reductase	Trimethoprim	99.80	99.80
sul2	Sulfonamide	Sulfamethoxazole	100.00	100.00
fosA6	Fosfomycin transferase	Fosfomycin	97.00	98.81
blaCTX-M-14 (dup)	ESBL	Cephalosporins	100.00	100.00
blaTEM-1B (dup)	β-lactamase	Penicillins	100.00	99.88
blaTEM-1B (dup2)	β-lactamase	Penicillins	100.00	99.88

4.7 Antimicrobial Resistance Gene Profiling – United States

ABRicate analysis of the U.S. *K. pneumoniae* genome (GCF_000597905.1_ASM59790v1) identified **6 resistance gene hits**, the smallest resistome of the three isolates. The key carbapenemase gene was **blaKPC-3**, a close variant of KPC-2 commonly associated with the globally disseminated ST258 lineage, detected on contig NZ_CP006919.1 with 100% coverage and identity. Additional genes included blaSHV-182, oqxA and oqxB (quinolone efflux), aac(6')-Ib (aminoglycoside resistance), and fosA6 (fosfomycin resistance). The more focused resistome of the U.S. isolate may reflect the distinct clonal background, local antibiotic selection pressures, and plasmid content characteristic of U.S. CRKP strains.



Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 1
#FILE	SEQUENCE	START	END	STRAND	GENE	COVERAGE	COVERAGE_MAP	GAPS	%COVERA
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006918.1	1270831	1273983	-	oqxB_1	1-3153/3153	===============	0/0	100
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006918.1	1274007	1275182	-	oqxA_1	1-1176/1176	===============	0/0	100
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006918.1	2787180	2788040	+	blaSHV-182_1	1-861/861	===============	0/0	100
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006918.1	4654753	4655172	-	fosA6_1	1-420/433	===============	0/0	97
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006919.1	8101	8982	+	blaKPC-3_1	1-882/882	===============	0/0	100
usa_GCF_000597905.1_ASM59790v1_genomic.fna	NZ_CP006919.1	23776	24381	-	aac(6')-Ib_1	1-606/606	===============	0/0	100

Figure 11. ABRicate Output for U.S. K. pneumoniae (Dataset 28): Six resistance gene hits showing oqxB, oqxA, blaSHV-182, fosA6, blaKPC-3, and aac(6')-Ib. The U.S. isolate displays the most compact resistome among the three isolates studied.

Table 6. Antimicrobial Resistance Genes Detected in the U.S. K. pneumoniae Isolate (ABRicate/ResFinder)

Gene	Class	Resistance Determinants	% Coverage	% Identity
blaKPC-3	Carbapenemase (Class A)	Carbapenems, β-lactams	100.00	100.00
blaSHV-182	β-lactamase	β-lactams	100.00	99.88
oqxA	Quinolone efflux	Nalidixic acid, Ciprofloxacin	100.00	100.00
oqxB	Quinolone efflux	Nalidixic acid, Ciprofloxacin	100.00	100.00
aac(6')-Ib	Aminoglycoside	Amikacin, Tobramycin	100.00	100.00
fosA6	Fosfomycin transferase	Fosfomycin	97.00	98.81

4.8 Comparative Resistome Analysis

Table 7 presents a comprehensive comparative summary of resistome profiles across the three isolates. The most striking finding is the geographic specificity of the dominant carbapenemase gene: blaNDM-1 in India, blaKPC-2 in Turkey, and blaKPC-3 in the United States. The Indian isolate exhibited the largest and most diverse resistome (23 genes), followed by Turkey (17 genes) and the United States (6 genes). Despite these differences, fosA6 (fosfomycin resistance) was detected in all three isolates, representing a conserved shared resistance determinant. Shared aminoglycoside resistance genes (aph(6)-Id, aph(3'')-Ib, aadA2) were present in India and Turkey but absent from the U.S. isolate. SHV-family β-lactamase genes were found in all three genomes, consistent with the intrinsic β-lactamase background of *K. pneumoniae*.

Table 7. Comparative Resistome Profile Summary Across Three Geographic Isolates

Feature	India (BAA-2146)	Turkey (ASM24018)	USA (ASM59790)
Primary Carbapenemase	blaNDM-1	blaKPC-2	blaKPC-3
Total Resistance Genes	23	17	6
ESBL Genes	blaCTX-M-15, blaCMY-6	blaCTX-M-14	None detected
Aminoglycoside Genes	rmtC, aac(3)-IIa, aac(6')-Ib, aph(3")-Ib, aph(6)-Id, aadA2	rmtB, aac(3)-IId, aph(3")-Ib, aph(6)-Id, aadA2	aac(6')-Ib
Quinolone Resistance	oqxA, oqxB, qnrB9, aac(6')-Ib-cr	None	oqxA, oqxB
Tetracycline Resistance	tet(A)	tet(G)	None
Fosfomycin Resistance	fosA6	fosA6	fosA6
Sulfonamide Resistance	sul1, sul2	sul2	None
Trimethoprim Resistance	dfrA14	dfrA12	None

Phenicol Resistance	None	floR	None
Resistome Complexity	Very High (MDR)	High (MDR)	Moderate

5. DISCUSSION

5.1 Quality Control and Read Preprocessing

The FastQC analysis of raw reads confirmed standard Illumina sequencing characteristics, with high per-sequence quality scores but reduced quality at the 3' ends of reads, adapter contamination, and expected sequence duplication levels. The total of 3,016,254 read pairs (907.8 Mbp per file) represented approximately 170× theoretical coverage of the ~5.3 Mbp *K. pneumoniae* genome, providing substantial sequencing depth. Trimmomatic preprocessing resolved the adapter contamination and low-quality tail issues while retaining 98.8% of reads (2,981,740 pairs), confirming efficient and stringent quality trimming without excessive data loss.

5.2 Genomic Diversity of the Indian Isolate

The detection of 30,261 genomic variants in *K. pneumoniae* BAA-2146 relative to the MGH 78578 reference genome underscores the substantial genomic divergence between these strains. The dominance of SNPs (85.77%) is consistent with the typical variant landscape observed in comparative genomic studies of *K. pneumoniae*. The 3,852 MNPs likely reflect complex substitution events accumulated over evolutionary time, while the 358 indels represent insertions and deletions potentially affecting gene function and regulatory sequences. The high variant density indicates that BAA-2146 and MGH 78578 represent evolutionarily distinct lineages, consistent with the clinical and epidemiological differences observed between NDM-1-positive South Asian strains and the North American reference strain.

5.3 Resistome Complexity of the Indian Isolate

The 23 resistance genes detected in the Indian isolate illustrate the characteristic multidrug resistance profile of NDM-1-positive *K. pneumoniae* strains from South Asia. The co-occurrence of blaNDM-1 with multiple other β-lactamase genes (blaCTX-M-15, blaTEM-1B, blaOXA-1, blaCMY-6, blaSHV-187) and resistance genes targeting quinolones, aminoglycosides, tetracyclines, sulfonamides, trimethoprim, and fosfomycin is consistent with the extensively drug-resistant (XDR) phenotype expected for this reference strain. The detection of rmtC, a 16S rRNA methyltransferase conferring pan-aminoglycoside resistance, is particularly noteworthy as it eliminates aminoglycosides as treatment options. These findings align with published reports characterizing BAA-2146 as a well-characterized NDM-1-positive multidrug-resistant reference strain.

5.4 Geographic Variation in Carbapenemase Genes

One of the most important findings of this study is the clear geographic specificity of dominant carbapenemase genes: blaNDM-1 in India, blaKPC-2 in Turkey, and blaKPC-3 in the United States. This result is consistent with established global epidemiology of carbapenem resistance, where NDM-type enzymes predominate in South Asia, KPC enzymes are widely prevalent in the United States, and KPC-2 is increasingly reported in Turkey and other regions. The Turkish isolate's blaKPC-2 carriage indicates the spread of KPC-type carbapenemases into the Mediterranean/Eurasian region, while the U.S. isolate's blaKPC-3 is characteristic of the ST258 high-risk clone that has disseminated widely in North America and Europe.

5.5 Differences in Resistome Size

The marked differences in resistome size (India: 23, Turkey: 17, USA: 6) likely reflect a combination of factors: clonal background and evolutionary history, local antibiotic selective pressures, plasmid content and mobile element carriage, and healthcare system antibiotic usage patterns. The more expansive Indian resistome may indicate a higher degree of plasmid-mediated gene acquisition driven by intensive antibiotic use in healthcare settings where many different antibiotic classes are employed. The relatively compact U.S. resistome, despite harboring a clinically critical KPC-3 gene, suggests that the ST258 lineage may have been selected primarily by carbapenem pressure rather than broad-spectrum multidrug selection.

5.6 Shared Resistance Determinants

The detection of fosA6 in all three isolates indicates its conservation as a common *K. pneumoniae* resistance determinant that has spread across different geographic lineages, possibly through horizontal gene transfer. The presence of oqxAB quinolone efflux genes in both the Indian and U.S. isolates but not in Turkey is notable and may reflect different plasmid backgrounds. The SHV-family β-lactamases detected in all three isolates are consistent with the intrinsic chromosomal SHV gene characteristic of *K. pneumoniae*

5.7 Significance of the Galaxy Platform

This study demonstrates the effectiveness of the Galaxy platform for conducting a comprehensive, reproducible NGS-based AMR analysis without requiring command-line expertise. The complete workflow, from raw reads to variant calling and resistance gene profiling, was executed within a unified environment with full parameter recording and history retention. This approach is particularly valuable for academic training in bioinformatics and genomic surveillance research.

5.8 Limitations

Several limitations should be acknowledged: only one representative isolate from each country was analyzed; functional annotation of variants was not completed; phenotypic antibiotic susceptibility data were not integrated; detailed plasmid content analysis (PlasmidFinder) and sequence typing (MLST) were not performed; and phylogenetic analysis was beyond the scope of this study. Despite these limitations, the study successfully demonstrates a robust comparative genomics framework.

6. CONCLUSION

The present study successfully demonstrated the application of whole genome sequencing and comparative genomics for investigating antimicrobial resistance in carbapenem-resistant *Klebsiella pneumoniae* isolates from India, Turkey, and the United States using the Galaxy bioinformatics platform.

Analysis of the Indian isolate (*K. pneumoniae* ATCC BAA-2146) revealed **30,261 total genomic variants** relative to the MGH 78578 reference genome, comprising **25,955 SNPs**, **3,852 MNPs**, and **358 indels**, demonstrating substantial evolutionary divergence and genomic plasticity.

Resistance gene profiling by ABRicate identified **23 resistance gene hits** in the Indian isolate, with **blaNDM-1** as the dominant carbapenemase alongside a highly complex resistome spanning β-lactamases, aminoglycosides, quinolones, tetracyclines, sulfonamides, trimethoprim, and fosfomycin. The Turkish isolate carried **17 genes** dominated by **blaKPC-2**, and the U.S. isolate harbored **6 genes** centered on **blaKPC-3**.

These findings confirm clear geographic variation in dominant carbapenemase genes and overall resistome complexity, with fosA6 emerging as a shared resistance determinant across all three isolates. The study highlights the power of open-source bioinformatics tools and the Galaxy platform for comprehensive, reproducible AMR genomic surveillance research.

7. FUTURE PROSPECTS

- Functional annotation of detected variants using SnpEff to predict the impact of SNPs and indels on gene function.
- Plasmid analysis using PlasmidFinder to determine whether resistance genes such as blaNDM-1 and blaKPC are plasmid-borne and to characterize horizontal gene transfer dynamics.
- Sequence typing using MLST to identify high-risk clonal lineages (e.g., ST258, ST11) associated with global dissemination.
- Phylogenomic analysis incorporating larger collections of CRKP isolates from multiple countries to reconstruct evolutionary relationships and identify transmission events.
- Integration with phenotypic antibiotic susceptibility testing data to enable genotype–phenotype correlation.
- Expansion to population-level analyses of dozens to hundreds of isolates for robust statistical inference about global resistance patterns.
- Application of long-read sequencing (Oxford Nanopore or PacBio) to resolve complete plasmid assemblies and mobile genetic element contexts.

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